

BENCHMARK: NON-DETECTION

Results

Mid-Transit Time (Tmid)	2458500.7795 +/- 0.0022 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.141 +/- 0.051
Transit depth (Rp/Rs)^2	1.99 +/- 1.45 [%]
Orbital Inclination (inc)	85.01 +/- 1.72
Airmass coefficient 1 (a1)	1.26 +/- 0.86
Airmass coefficient 2 (a2)	-0.03 +/- 0.14
Scatter in the residuals of the lightcurve fit is	63.21 %
Best Comparison Star	None
Optimal Aperture	2.64
Optimal Annulus	5.0
Transit Duration (day)	0.043 +/- 0.038

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.141 ± 0.051 vs 0.131 (deviation 0.2σ)
Duration fit vs published	0.043 d vs 0.1075 d (deviation 1.7σ)
Mid-time O–C	0.8 min
Depth SNR	2.8
Residual scatter	63.21 %

Light curve

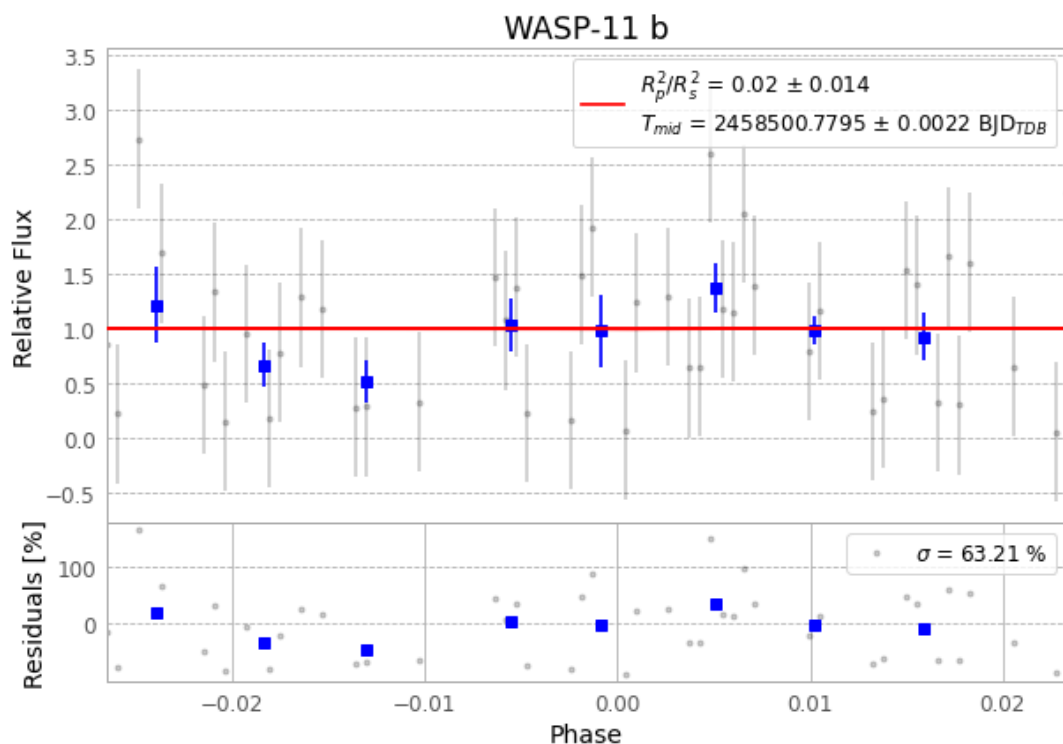


Figure 1 — FinalLightCurve_WASP-11 b_2019-01-16.png (SHA-256 76d371bc37736575...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [449, 472], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 7f21374f4faac8e5...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit ba3ef8b

Data provenance

93 FITS frames (61 MB), HATP-10190117040912.FITS → HATP-10190117084512.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-HATP-10-190117.log (0d7762e60536...), FinalLightCurve_WASP-11 b_2019-01-16.png (76d371bc3773...), FinalLightCurve_WASP-11 b_2019-01-16.csv (b6859dfc70e2...)

Compute resources

Wall time 120.2 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-07-31 07:07Z.